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Unit “ 4 : Session “ 12 : SLO “ 1 | Influence on Instruction Sets “ Choose the Best Answer

■ I. Multiple Choice

1. b) Simplified, single-cycle instructions “ RISC design is driven by simple instructions that execute in one cycle.
2. a) Easier pipelining and decoding “ fixed-length instructions align fetch/decode stages neatly.
3. c) Memory-to-memory operations “ CISC instructions can operate directly on memory operands.
4. b) SIMD “ one instruction processing multiple data elements boosts modern performance.
5. a) Code density “ embedded systems have little memory, so compact code matters most.
6. a) To reduce branch penalties “ predication executes both paths conditionally, eliminating branches.
7. b) Backward compatibility “ x86 kept variable-length instructions to run legacy software.
8. c) More compact code “ register-memory instructions do more per instruction, shrinking code size.
9. b) Out-of-order execution “ dynamic scheduling most reshaped modern ISA/microarchitecture design.
10. a) Increased compiler complexity vs. code flexibility “ more addressing modes give flexibility but complicate the compiler and hardware.